# COMP396 Weekly Meeting Minutes

# (Week 4)

**Date & Time:** Wednesday, October 15th, 2025, 12:00-13:00

**Location:** Electrical Engineering Building, Lecture Room E4

**Note Taker:** Xue Yudan

**Attendees:** Xue Yudan, Zhang Olivia, Zhao Saria, Wong Haw

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***Note:*** *This meeting was rescheduled from the originally planned time on Friday, October 17th, as Zhao Saria has GP appointments on October 17th for a lower back injury.*

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## 1. Agenda:

- Check the completion status of last week's tasks(C1-C3), including the initial strategy conception and chart application.
- Share the observations from the analyses of ten original data (released on 1st Oct) and discuss their impact on the choice of strategies.
- Confirm the core strategic direction and the basic idea for the investment portfolio in the first phase, based on the data analysis.
- Organize and allocate the core tasks for next week, shifting the focus primarily to the coding implementation of the strategies and report preparation.

## 2. Discussion & Decisions:

### 2.1. Task Review and Result Confirmation

Our team confirms that all tasks assigned during last week's meeting (Week 3) have been completed successfully.

- **Initial Strategy Conception (C1):** All team members have completed the initial strategy conception documentation for the three independent strategies they are responsible for.
- **Command of Chart Analysis (C2):** Team members have gained a deep understanding of the composition and information interpretation methods of key charts by learning the code from Lecture 6, which lays the foundation for data-driven decisions.
- **Team management (C3):** All team members have organized the issues encountered in the current phase and successfully scheduled a meeting with the Supervisor.

## 2.2. Data Analysis Findings & Strategy Direction

After mastering data visualization skills, Yixuan Zhao conducted a preliminary analysis of the ten asset data sets provided on Canvas and shared the key findings with other members:

- **Market Trend Divergence:** Asset 04 exhibits a significant long-term upward trend.
- **General Downward Trend:** Most of the remaining assets (excluding 04 and 06-09) show varying degrees of a downward trend.
- **Range-Bound/Sideways Market:**Assets 06, 07, 08, and 09 exhibit range-bound characteristics and are essentially trend-neutral.

Following the analysis, our team thoroughly discussed and initially set the core strategy for Phase One: to concentrate on shorting assets with significant downtrends, while temporarily excluding the trend-neutral, range-bound assets to enhance the strategy's focus and effectiveness.

## 2.3. Next Phase Plan: From Ideation to Implementation

In order to have a productive discussion with the Supervisor, our team's focus will shift from theory to practice.

- **Core target：** Before next week’s meeting, finish the initial coding writing of three independent strategies.
- **Brainstorm of the investment portfolio strategies:**
    - Apply the three distinct strategies to three independent assets which have the clear downward trend.
    - When constructing the investment portfolio, we should take a **mean-reversion** strategy or a risk management model (such as one based on **Conditional Value at Risk**) into consideration, to achieve strategic complementarity and risk control.

## 3. Task Allocation:

To ensure the successful achievement of the next phase's objective, our team has made the following task allocation:

| ID | Module | Task Description | Owner | Due Date |
|---|---|---|---|---|
| **D1** | Strategy Implementation | Complete the initial Python coding for the three independent strategies based on the core 'short-falling-assets' policy. Ensure the logic is clear and executable in preparation for the meeting with the Supervisor. | ZS (Main)<br>XY & WH & ZO (Assist) | Oct. 22 |
| **D2** | Visualization & Analysis Application | Using AI tools (e.g., ChatGPT) to generate simulated financial data, applied the charting techniques learned in L6 to visually analyze and independently interpret the results, and then used AI to verify the accuracy. The goal is to internalize chart comprehension into a proficient, applied skill. (Detailed plan available in the Appendix) | All Members | Oct. 24 |
| **D3** | Advanced Concepts & Report Preparation | **Task 1:** Re-study the content on Overfitting in L7 and consider how to effectively avoid this issue in the middle and later stages of the project (Part 2 & 3).<br><br>**Task 2:** Deeply study the requirements and details of each section of the Design Document and collaboratively create a | WH & XY & ZO | Oct. 24 |

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|---|---|---|---|---|
| | | concise and clear overall outline of the report structure for all members' reference. | | |

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## 4. Current Challenges and Issues

- **The Shift from Theory to Practice:** The primary challenge has now shifted from 'what strategy to choose' to 'how to implement the strategy efficiently and accurately.' Ensuring the robustness and logical rigor of the strategy code, as well as integrating the three independent strategies into a single effective investment portfolio, will be the upcoming technical difficulties.

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## 5. Supervisor Question

- We intend to finalize the core briefing materials for the Supervisor meeting by next Wednesday. These materials will contain the analysis of the preliminary data and generated charts, a confirmed strategic direction, and the near-complete initial draft of the strategy code. The meeting aims to obtain Supervisor’s feedback on the technical implementation and the portfolio construction methodology.

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## 6. Next Meeting Arrangement

- **Date & Time:** Thursday, October 23th, 2025, 15:00-17:00 (After the meeting with Supervisor)
- **Location:** Sydney Jones Library / Abercromby Study Room 1
- **Key Agenda Items:**
  - Code Review of Individually Completed Strategies (D1).
  - Share Progress on Chart Application Practice (D2).

- Organize and summarize all advice provided by the Supervisor and formulate a strategy modification plan based on the feedback.

## Appendix:

### Deep Learning Task Allocation for Charts and Indicators

### (Task D2 Addendum)

To systematically master all visualization and analysis tools from Lecture 6, our team has decided to divide the **charts** and related **indicators** into three major modules, with a dedicated person assigned to lead the in-depth study and practice for each. Each leader must, before next week's meeting, achieve proficiency in the creation methods, data sources, and the core information revealed by their assigned content.

#### 1. Investment Portfolio Macro Performance Assessment (Lead: Zhang Olivia)

- Core Task: Focus on assessing the overall macro performance and basic returns of the strategy.
- Learning Charts**:**
  - Portfolio Cumulative PnL
  - Equity Curve
  - Realized PnL from Fills
- Learning indicators:
  - Profit and Loss, PnL
  - Simple Return
  - Log Return

#### 2. Investment Portfolio Risk and Detailed Analysis (Xue Yudan)

- Core Task: Focus on assessing the strategy's risk, drawdowns, and the detailed performance of its constituent series.
- Learning Charts:
  - Underwater/Drawdown Plot
  - Per-series Cumulative PnL

    - Price, Position & PnL
- Learning indicators:
    - Drawdown / Max Drawdown

## 3. Market Data and Price Analysis (Zhao Saria)

- Core Task: Focus on single-asset data exploration, price behavior, and the superimposed analysis of common technical indicators.
- Learning Charts:
    - Price and Return Plots
    - Candlestick Chart and Volume
    - Indicators & Overlays
    - Histogram
    - Small Multiples / Seaborn Facets
- Learning indicators:
    - Moving Average, MA/SMA
    - Bollinger Bands
    - Standard Deviation
    - Volume

## 4. Multidimensional Relationships and Parameter Optimization (Wong Haw)

- Core Task: Focus on analyzing complex relationships between multiple variables (including price and other indicators) and the visualization of strategy parameters.
- Learning Charts:
    - Correlation Heatmap
    - Pairplot for Nonlinear Relations
    - Fitness Landscape for Parameter Optimization
- Learning indicators:
    - The focus is on applying all the above indicators for multidimensional analysis, for example, analyzing the relationships between price, moving averages, and the upper and lower bands of the Bollinger Bands in a correlation heatmap and a pair plot.